Barathi Ganesh Hullathy Balakrishnan (Barathi Ganesh HB)

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Research Vision

Doctoral Researcher investigating unified **Multimodal, Multilingual, and Multitask (M3) Language Models (M3LM)**, bridging a decade of production AI engineering with foundational speech-text representation learning. My research resolves the topological conflict between continuous acoustic paralinguistics and discrete semantic tokens across **100+ global and underrepresented languages**, and ensuring AI systems to serve diverse linguistic identities with **demographic fairness, structural equity, and hardware-level efficiency**:

- **Completed-** ***Multi-View Auditory Perceiver & Subspace Disentanglement (M3LM-SpeechEncoder):*** Designed and empirically validated a two-stage, dual-stream speech foundation architecture (Wav2Vec2-BERT 2.0 + ReDimNet-B3) coupled with the dynamic **Parallel Omni Adapter**, asymmetric stop-gradient gated cross-modal fusion, curriculum gated homoscedastic uncertainty weighting, and all-to-all Gradient Reversal Layer (GRL) adversarial confounder pruning. Compresses 50 Hz audio sequences by 8x into 6.25 Hz semantic soft-prompts plus 4 disentangled 512-D biometric prefix tokens, cutting quadratic LLM self-attention FLOPs by 61.7x - 63.7x and KV-cache memory by 87.1%.
- **Ongoing-** ***Text-LLM Bridging & M3LM-Corpus Curation:*** Interfacing autoregressive Text LLMs (Gemma-4 / Qwen2.5) directly with the frozen M3LM-SpeechEncoder dual-prefix bridge (compressing frame-level phonetic tokens alongside explicit language, speaker, gender, and age vectors via MLP-targeted LoRA without Per-Layer Embedding gradient isolation). In parallel constructing the **M3LM-Corpus** to address the critical lack of a demographically balanced, semantically validated multimodal, multilingual, and multitask dataset across 100+ languages through weak supervision pseudo-labeling, cross-modal semantic embedding validation, NeMo Forced Aligner, and zero-leakage duration-weighted greedy stratification.
- **Next-** ***End-to-End M3LM Training on Distributed Hardware:*** Scaling and training the complete, architecturally proven **M3LM** generative speech-text foundation model on the curated **M3LM-Corpus** using a multi-node GPU cluster, establishing an open, bias-audited omni-modal model capable of sub-1-second multilingual speech recognition, translation, paralinguistic reasoning, and spoken dialogue across 100+ global and underrepresented languages.

Education

- **Ph.D. in Information and Communication Engineering (Specialization: Multimodal AI)** | Kitami Institute of Technology, Hokkaido, Japan *(Oct 2025 - Expected March 2029)*
    - **Laboratory:** Text Information Processing and Informatics Laboratory (TIP Lab) | **Advisor:** Prof. Michal Ptaszynski
    - **Academic System:** Credit-Based Doctoral Coursework & Research Progression (Completed 1st Year; Full-Time Doctoral Candidate)
    - **Dissertation Focus:** *M3LM: Bias-Aware Multimodal, Multilingual, and Multitask Language Models and Quantitatively Validated Open Corpora*
- **Master of Technology (M.Tech) in Computational Engineering and Networking (AI Specialization)** | Amrita Vishwa Vidyapeetham, Coimbatore, India *(Sep 2013 - Aug 2015)*
    - **Academic Standing:** First Class | **Thesis Advisor:** Dr. Anand Kumar M and Dr. Soman KP
    - **Focus:** Natural Language Representation and Processing, Distributional & Distributed Vector Space Models of Semantics
- **Bachelor of Engineering (B.E.) in Electronics and Communication Engineering** | CSI College of Engineering, Anna University, Chennai, India *(Sep 2009 - Apr 2013)*
    - **Academic Standing:** First Class
    - **Focus:** Digital Signal and Image Processing

Academic Research Experience

Doctoral Researcher | Text Information Processing & Informatics Lab (TIP Lab), Kitami Institute of Technology, Japan *(Oct 2025 - Present)*

- **Proving the M3LM-SpeechEncoder & Parallel Omni-Adapter Architecture:** Designed and trained a two-stage, dual-stream foundation encoder across 40+ languages and 3,666 speakers. Coupled a 24-layer continuous semantic conformal backbone with a 2D/1D spectral-temporal biometric backbone via asymmetric stop-gradient gated cross-modal fusion, 25-state task routing, Conformer biometric/ASR adapters, and dynamic convergence-based task shedding, achieving 7.98x faster speaker adaptation, 99.24% closed-set speaker accuracy (99.22% BAcc, 0.344% EER), 1.876% open-set EER on TidyVoiceX Dev (36.84% relative reduction over VoxBlink2/VoxCeleb2-pretrained SimAM-ResNet34 cosine baselines [2.970%] and 13.94% lower than full NAP+AS-Norm [2.180%] without back-end score normalization), 99.54% zero-shot Gender balanced accuracy (2.00% open-set EER) on VoxCeleb1-O, 79.44% (Test) / 74.03% (Dev) adjacent-bin (5-year) age accuracy, 16.36% greedy CTC normalized CER, and 89.45% zero-shot FLEURS Language Identification balanced accuracy (91.17% Macro-F1; +17.82% over w2v-BERT 2.0 and +15.92% over mSLAM baselines).
- **High-Throughput LLM Perceptual Interface:** Integrated a strided 1D convolutional context compressor (6.25 Hz frame rate) paired with 4 disentangled 512-D paralinguistic prefix vectors (speaker, language, gender, age), reducing downstream autoregressive Text-LLM quadratic self-attention compute by **61.7x-63.7x** and KV-cache footprint by **87.1%** while sustaining **34.7x-37.3x faster-than-real-time** inference across 99,265 benchmark utterances.
- **Acoustic-Unconstrained Speech Translation (AURA-ST):** Built a 3-stage speech-to-text translation pipeline projecting dual-stream semantic and paralinguistic features through a 4x convolutional temporal subsampler into a frozen Gemma-4-E2B LLM; diagnosed and resolved Per-Layer Embedding (PLE) gradient isolation using MLP-targeted LoRA to reach **91.29 validation SacreBLEU** on low-resource African languages (Hausa, Igbo, Yoruba).

- **Global M3LM Multimodal Corpus & Quantitative Semantic Validation:** Working on the aggregation, weak-supervision pseudo-labeling, and quantitative semantic-embedding validation of a massive, multilingual speech-text corpus spanning **100+ global languages** (integrating 30+ open datasets across ASR, S2TT, TTS, LID, SID, emotion, sentiment, and dialect tasks) to train the full generative M3LM foundation model.

Research Associate | Center for Computational Engineering & Networking (CEN), Amrita Vishwa Vidyapeetham, India *(Jul 2017 - Mar 2018)*

- **Distributional & Semantic Vector Representations:** Developed text representation architectures (Vector Space Models of Semantics, Doc2Vec, TDNN, Bi-LSTM, and CRF sequence models) for mixed-script information retrieval, code-mixed entity linking, and clinical/legal information extraction.
- **Benchmarking & Shared Task Leadership:** Co-organized international evaluation campaigns at FIRE (INLI, DPIL, MTIL) and mentored research groups in sequence labeling and multilingual NLP.

Industry R&D Leadership Experience (10 Years)

Head of Product Development & R&D | RBG.AI, Coimbatore, India *(Aug 2020 - Oct 2025)*

- **Machine Learning Service Stack (MLStack):** Directed the AI R&D organization in architecting and deploying an enterprise-grade, microservice-orchestrated multimodal AI platform structuring complex unstructured information across **Voice, Text, and Vision** streams.
- **Low-Resource Speech & Dialect Systems:** Led applied research on parameter-efficient Whisper adaptation (WhispAdapt), low-resource speech corpus creation (CoRePooL), post-call speech analytics (speaker diarization, emotion, sentiment, and abuse extraction), and 4-bit quantized multilingual translation for consumer GPU deployment.
- **Cross-Sector Production Deployment:** Delivered latency-bounded AI pipelines across BFSI (automated deal-word confirmation, anomaly detection), Healthcare (ASR/OCR medication auto-completion, clinical knowledge graphs, radiology reporting), Automotive (voice infotainment, principal speaker recognition), and Manufacturing/AEC (2D/3D layout extraction, CAPP feature recognition).

Chief Technology Officer (CTO) & R&D Product Manager | Arnekt Solutions Pvt. Ltd., Pune, India *(Mar 2018 - Jun 2020)*

- **Cognitive Intelligence as a Service (CIaaS):** Architected a distributed, parallelized microservice platform serving production NLP, ASR, and Computer Vision models for enterprise digital transformation.
- **Conversational AI & Autonomous Systems:** Built a low-cognitive-load Conversational Agent Platform for SMEs (co-organizing the Arnekt-IECSIL@FIRE-2018 shared task across 5 Indian languages) and engineered an end-to-end autonomous driving ML pipeline using imitation and reinforcement learning in CARLA and AirSim simulation environments.

Research Scientist Analyst - Team Lead | Accenture Innovation Center, Bengaluru, India *(Dec 2016 - Jul 2017)*

- **Enterprise Multilingual Automation:** Led an applied research team automating end-to-end enterprise procurement workflows by combining Neural Machine Translation (NMT), Named Entity/Information Extraction, and OCR.

AI & Cognitive Science Practitioner | Tata Consultancy Services (AI Practice Lab), Kochi, India *(Mar 2015 - Dec 2016)*

- **Foundational NLP & Vision Engines:** Engineered reusable coreference resolution, sentence simplification, document classification, and OCR modules deployed across enterprise Healthcare and BFSI systems.

Technical Expertise

- **Frameworks & Toolkits:** PyTorch, Accelerate, Transformers, NeMo Speech, FairSeq, SpeechBrain, Gemma-4, LLaMA, Whisper, RE-USE, XLM-RoBERTa, MADLAD-400, NLLB-200, Ray, MLflow, Optuna, Docker, Kubernetes, FastAPI, Python, R, MATLAB.
- **Multimodal & Speech-Language Architectures:** Unified Speech-Text Transformers, Multi-View Dual-Stream Fusion, Asymmetric Stop-Gradient Cross-Attention, Gumbel-Softmax Product-Codebook Vector Quantization (92.8% codebook utilization, 98.7% Shannon coding efficiency), Parallel Omni-Adapters, Non-Autoregressive CTC Decoding, Acoustic-LLM Prefix Projection, MLP-Targeted LoRA / QLoRA, 4-Bit Outlier-Aware Quantization.
- **NVIDIA Hardware & Distributed Compute Stack:** 2-Node DGX Spark Cluster, RTX A6000 (48 GB VRAM), RTX 4090, TensorRT, Triton Inference, ONNX Runtime, vLLM, Automatic Mixed Precision (AMP) with Dynamic Gradient Scaling, Pinned-Memory Asynchronous Multi-Worker Data Pipelines, Native PCM Bitstream Ingestion, Dynamic Length-Grouped Sampler.
- **Fairness, Disentanglement & Data Curation:** All-to-All Adversarial GRL Confounder Pruning, Curriculum-Gated Homoscedastic Task Uncertainty Weighting, Additive Angular Margin Softmax (AAM-Softmax), Zero-Leakage Duration-Weighted Greedy Intersectional Stratification, Weak Supervision & Pseudo-Labeling, NeMo Forced Aligner, Semantic Embedding Validation (Margin-Based LID, CSLS).

Selected Publications (60+ Total Publications | h-index: 13)

Published & Accepted Papers (Recent Focus: 2024-2026)

1. **Barathi Ganesh HB**, Michal Ptaszynski, Meenakshi, and Jairam R. (2026). **TIP-RBG-AI: Overcoming Orthographic Discrepancies via Algorithmic Translinear Pipelines and Phylogenetic Script Mapping**. *Published in Proceedings of EMNLP 2026 (WMT26 Indic-MT Shared Task)*, ACL. *(Resolved cross-script orthographic fragmentation across low-resource languages via algorithmic translinearization and phylogenetic script mapping).*
2. **Barathi Ganesh HB**, Michal Ptaszynski, Jairam R, and Reshma Unnikrishnan. (2026). **AURA-ST: Acoustic-Unconstrained Residual Architecture for Speech Translation**. *Proceedings of the 23rd International Conference on Spoken Language Translation (IWSLT 2026)*, ACL, pp. 247-254, San Diego, USA. *(Introduced a 3-stage dual-stream semantic/paralinguistic encoder + 4x convolutional subsampler projecting audio tokens directly into a frozen Gemma-4-E2B backbone via MLP-targeted LoRA; resolved Gemma-4 Per-Layer Embedding gradient isolation and achieved 91.29 validation SacreBLEU across Hausa, Igbo, and Yoruba).*
3. **Barathi Ganesh HB**, Michal Ptaszynski, Rene Melendez, and Juuso Eronen. (2026). **KIT-TIP-NLP at MultiPride: Continual Learning with Multilingual Foundation Model**. *Proceedings of the Ninth Evaluation Campaign of Natural Language Processing and Speech Tools for Italian (EVALITA 2026)*, CEUR Workshop Proceedings, pp. 302-314, Bari, Italy. *(Developed a continual MLM + inductive transfer learning pipeline on XLM-RoBERTa with 3x cross-lingual back-translation augmentation and ROC-optimized language-specific decision thresholds, yielding 2-5% absolute Macro-F1 gains without retraining across English, Spanish, and Italian demographic/reclamatory contexts).*

4. **Barathi Ganesh HB** and Michal Ptaszynski. (2025). **RBG-AI: Benefits of Multilingual Language Models for Low-Resource Languages**. *Proceedings of the Tenth Conference on Machine Translation (WMT 2025 / EMNLP 2025)*, ACL, pp. 1233-1239, Suzhou, China. *(Evaluated zero-shot cross-lingual transfer across Tibeto-Burman, Indo-Aryan, and Austroasiatic languages using MADLAD-400, NLLB-200, and mBART-50; implemented outlier-aware 4-bit quantization for a 4x memory footprint reduction on consumer GPUs while uncovering script-vs-typology transfer asymmetries).*
5. Jairam R, Premjith B, and **Barathi Ganesh HB**. (2025). **WhispAdapt: Multilingual Dialect Recognition in Low-Resource Dravidian Languages**. *Array (Elsevier)*, 2025, 100629. *(Adapted Whisper encoders via LoRA and QLoRA across a newly curated 65.9-hour 12-dialect Dravidian speech corpus, achieving 90% Macro-F1 on Tamil, 93% on Kannada, and 97% on Malayalam outperforming Wav2Vec2, HuBERT, and WavLM by up to 25% and generalizing zero-shot to unseen accents).*
6. **Barathi Ganesh HB**, Premjith B, Jyothish Lal G, et al. (2024). **CoRePooL-Corpus for Resource-Poor Languages: Badaga Speech Corpus**. In *Automatic Speech Recognition and Translation for Low Resource Languages*, Wiley-IEEE Press, pp. 193-211. *(Released 420 minutes of multi-task supervised speech data [ASR, TTS, translation, gender, speaker ID] and 968 minutes of unsupervised multilingual speech for the under-resourced Badaga language alongside open-source foundation model baselines).*
7. **Barathi Ganesh HB**, Jairam R, Michal Ptaszynski, Reshma U, Jyothish Lal G, and Premjith B. (2026). **DisentangLID: A Robust Encoder with Dynamic Adapter for Language Identification**. *Accepted at The Speaker and Language Recognition Workshop (TidyLang / Odyssey 2026; non-archival / opted out of registration).*

Journal Submissions & Preprints (2025-2026)

8. **Barathi Ganesh HB**, Michal Ptaszynski, Reshma Unnikrishnan, and Jairam R. (2026). **M3LM-SpeechEncoder: Multi-View Acoustic and Semantic Fusion for Foundation Models**. *Submitted to Computational Linguistics (ACL / MIT Press).* (*Proves the two-stage dual-stream foundation encoder and Parallel Omni-Adapter with 25-state routing, Conformer adapters, dynamic task shedding, and GRL adversarial confounder pruning; achieves 0.344% in-domain EER, 1.876% open-set EER on TidyVoiceX Dev, 89.45% zero-shot FLEURS LID BAcc [91.17% Macro-F1], 99.54% zero-shot VoxCeleb1-O gender BAcc, 16.36% greedy CTC CER, 61.7x-63.7x LLM self-attention FLOP reduction, and 37.3x real-time throughput.)*
9. **Barathi Ganesh HB**, Michal Ptaszynski, Reshma U, Jairam R, and Meenakshi Sharma. (2026). **A Systematic Survey of Modality Fusion Strategies in Multimodal Multilingual Multitask Speech-Text Transformers**. *SSRN Preprint 7615598 / Under Review at Information Fusion (Elsevier).*
10. **Barathi Ganesh HB**, et al. (2025). **Cascaded-ASR: A Scalable Multilingual Speech Recognizer for Tamil and English Languages**. *SPELL 2025 (Scalable ASR architecture robust with noise variations and domain adaptation).*

Selected Earlier Foundational & Shared Task Overview Publications (2015-2020)

- **Barathi Ganesh HB**, K. P. Soman, Reshma U, et al. (2018). **Overview of Arnekt IECSIL at FIRE-2018 Track on Information Extraction for Conversational Systems in Indian Languages**. *Working Notes of FIRE 2018 (CEUR-WS).*
- **Barathi Ganesh HB**, Anand Kumar M, K. P. Soman, Paolo Rosso, et al. (2017, 2018). **Overview of the First and Second Shared Tasks on Indian Native Language Identification (INLI at FIRE-2017 & FIRE-2018)**. *CEUR Workshop Proceedings.*
- **Barathi Ganesh HB**, Anand Kumar M, and K. P. Soman. **From Vector Space Models to Vector Space Models of Semantics** *(1st Place in Mixed-Script Information Retrieval & Classification).*

- Additional 45+ peer-reviewed papers spanning Code-Mixed Entity Recognition (CRF/Bi-LSTM), TDNN-BiLSTM Ink Recognition, SemEval Affect/Emotion Analysis, and Distributional Semantics (full list on Google Scholar and Scopus).

Open-Source Corpora, Editorial Service & Community Leadership

- **M3LM Global Multimodal Corpus & Quantitative Validation Pipeline (Ongoing):** Architecting a unified, open-source multilingual speech-text foundation corpus spanning **100+ global languages** (50K-64K+ hours of curated speech and multi-trillion token text alignments):
    - **Integrated Speech, Paralinguistic & Multimodal Sources:** FLEURS, CoVoST 2, Common Voice, Facebook OmniASR / MMS, People's Speech, CVSS, TidyVoice, LibriSpeech, VoxPopuli, Multilingual LibriSpeech (MLS), Granary, YODAS, Libri-Light, IndicVoices, IndicVoices-R, Shrutilipi, Kathbath, Nirantar, MUCS, BPOC, Vaani / SPIRE, BhasaAnuvaad, CREMA-D, RAVDESS, and ESD (paralinguistic emotion).
    - **Integrated Multilingual Text, Translation & Sentiment Sources:** FLORES-200, Samanantar, Aksharantar, IndicSentiment, Tatoeba, mC4 / OSCAR, CulturaX, Nemotron-Pretraining-Indic-v1, Sangraha, FineWeb-2, Dolma, and Aya Collection.
    - **Quantitative Semantic Validation & Weak-Supervision Pseudo-Labeling:** Enforces a 3-stage quantitative validation pipeline using semantic and cross-modal embeddings (EmbeddingGemma 2, margin-based LID, Cross-Domain Similarity Local Scaling [CSLS], NeMo Forced Aligner mathematical CTC length, and CTC topological feasibility guards $T_{acoustic} \geq U_{target}$), populates missing task annotations via **dual-model weak supervision** (semantic transcripts + age-gender for continuous biological age and categorical gender), and partitions splits via **zero-leakage duration-weighted greedy intersectional stratification** across speaker, language, gender, and 5-year age cohorts.
- **CoRePooL Open-Source Initiative:** Principal architect of the Corpus for Resource-Poor Languages (Badaga speech corpus, 65.9-hour 12-dialect Dravidian corpus, and baseline checkpoints).
- **Journal Editorial Board & Peer Review Service:**
    - **Editorial Advisory Board Member:** *Information Processing & Management: Applications* (Elsevier).
    - **Active Peer Reviewer:** *Information Processing & Management (IPM)* (Elsevier, Q1) and *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)* (ACM).
- **International Shared Task Organizer & Advisory Committee Member (7+ Shared Tasks):**
    - **Organizer & Session Chair:** *Indian Native Language Identification (INLI)* at FIRE-2017 & FIRE-2018 (56 registered international teams).
    - **Lead Organizer:** *Information Extractor for Conversational Systems in Indian Languages (Arnekt-IECSIL 2018)* at FIRE-2018 (NER and Relation Extraction across Hindi, Kannada, Malayalam, Tamil, and Telugu).
    - **Advisory & Technical Committee Member:** *Detecting Paraphrases in Indian Languages (DPIL)* at FIRE-2016; *Machine Translation in Indian Languages (MTIL)*.
- **Technical Workshop Organizer & Benchmark Evaluations:** Organized 4 national/international workshops (*DeepChem*, *DeepSci*, *AISec*, and *Linear Algebra & Optimization for Signal Processing*) and competed across 20+ international shared tasks (WMT, IWSLT, Odyssey TidyLang, EVALITA, SemEval, FIRE, PAN-CLEF, #Micropost NEEL, SMM4H, IWSPA-AP, SAIL, CDMC).